

# KYLE J. KLUGEWICZ

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## EDUCATION

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| <b>Columbia University</b>                                                                                        | New York, NY      |
| <b>Master of Science in Electrical Engineering</b>                                                                | Expected Dec 2027 |
| Coursework: Deep Learning, Digital Signal Processing, Digital Integrated Circuits, Millimeter-Wave Circuit Design |                   |
| <b>University of Texas at Arlington</b>                                                                           | Arlington, TX     |
| <b>Bachelor of Science in Electrical Engineering, GPA: 3.7/4.0</b>                                                | Aug 2026          |

## RESEARCH EXPERIENCE

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| <b>University of Texas at Arlington (UTA)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Arlington, TX       |
| <b>Lead Student Researcher, Neural Decoding</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Apr 2026 – Present  |
| <ul style="list-style-type: none"><li>Led a study on causal finger-flexion decoding from electrocorticography (ECoG), training a 12,000-parameter temporal convolutional network (TCN) across three subjects of Brain-Computer Interface Competition IV Dataset 4.</li><li>Rebuilt training and evaluation pipelines to eliminate data leakage and match official competition test splits, lifting mean decoding correlation from 0.41 to 0.53.</li><li>Quantized weights to 8-bit integers and deployed through ExecuTorch on a Radxa Zero 3W board, holding correlation at 0.51 in a 37 KB model, smallest deep ECoG decoder reported on this benchmark.</li><li>Measured streaming inference at under 35 ms per window with zero deadline misses; authoring a first-author manuscript targeting IEEE Engineering in Medicine and Biology Conference (EMBC).</li></ul> |                     |
| <b>UT Arlington Research Institute (UTARI)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Fort Worth, TX      |
| <b>Research Assistant, Rehabilitation Robotics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jan 2026 – May 2026 |
| <ul style="list-style-type: none"><li>Built a graphical user interface (GUI) driving a hand, wrist, and elbow exoskeleton for stroke rehabilitation therapy.</li><li>Developed a control system for achieving finger actuation within 1° of accuracy.</li><li>Revised circuit schematics and backend software to broaden therapy routines and improve session logging for clinicians.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |

## PROFESSIONAL EXPERIENCE

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| <b>Misonix, Inc. (Bioventus)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Farmingdale, NY                                               |
| <b>Research and Development (R&amp;D) Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sep 2020 – Aug 2021, Dec 2021 – Jan 2022, May 2022 – Aug 2022 |
| <ul style="list-style-type: none"><li>Executed verification testing and data collection, including 510(k) submission testing, for ultrasonic surgical devices used in precision bone cutting.</li><li>Engineered a tube set for a BoneScalpel Access ultrasonic handpiece and authored a first sterile-field Instructions for Use (IFU), partnering with regulatory affairs and vendors to confirm autoclave compatibility.</li><li>Drove an engineering change order (ECO) for disposable packaging, releasing updated prints and auditing each vendor submission against specification.</li></ul> |                                                               |
| <b>Oncor Electric Delivery</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fort Worth, TX                                                |
| <b>Asset Management Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | May 2026 – Aug 2026                                           |
| <ul style="list-style-type: none"><li>Partnered with cross-functional teams to build a SQL-based tracking system for asset management and grid modernization.</li><li>Modeled a risk score in SQL using statistical analysis to rank distribution assets for inspection and replacement.</li></ul>                                                                                                                                                                                                                                                                                                  |                                                               |
| <b>POWER Engineers</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Fort Worth, TX                                                |
| <b>Distribution Design Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | May 2025 – Aug 2025                                           |
| <ul style="list-style-type: none"><li>Engineered overhead and underground distribution circuits, spanning double-circuit, armless, and underground residential layouts, to cut cost while raising reliability.</li><li>Calculated conductor sag, guy-wire loading, and cable pull to satisfy Texas Department of Transportation (TxDOT) rules, then delivered construction packages covering permits, bills of materials, and field-ready prints.</li></ul>                                                                                                                                         |                                                               |

## TECHNICAL PROJECTS

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| <b>Portable Electroencephalography (EEG) Headband, UTA Design Project</b>                                                                                                                                                                                                                                                                             | Aug 2025 – Aug 2026 |
| <ul style="list-style-type: none"><li>Built a low-noise analog front end with an instrumentation amplifier and cascaded Butterworth filters to isolate neural signals from environmental noise.</li><li>Programmed microcontroller firmware to sample EEG, filter in real time, and output control signals for hands-free device operation.</li></ul> |                     |

## TECHNICAL SKILLS

**Programming:** Python, C++, C, C#, MATLAB, SQL, JavaScript, HTML

**Machine Learning:** PyTorch, ExecuTorch, quantization-aware training, edge deployment, latency profiling

**Hardware and Tools:** LabVIEW, Multisim, LTspice, Simulink, AutoCAD, Autodesk Fusion, Ansys, Git, Visual Studio Code